Dx7

Canonical Data Model

Dialysis Scope — Korn Gallardo Build Reference

Document	Dx7 CDM — Dialysis Scope
Based On	SantéXA CDM v1.0 — February 2026
Version	Dx7 CDM v1.0
Date	March 2026
Scope	HL7 ingestion, normalization, and canonical result storage for Dx7 v1
Classification	CONFIDENTIAL — Authorized Distribution Only
Instruction	Build exactly as defined below. Do not invent mappings. Do not auto-guess.

1. Global Rules

The following rules are absolute and apply to every table, query, and operation.

#	Rule
1	All operational tables MUST include tenant_id directly — not derivable by FK chain alone.
2	No cross-tenant queries. Tenant isolation is absolute.
3	No automatic data purge. Retention is indefinite by default.
4	All business logic lives in the API layer. No business logic in the database. PostgreSQL RLS is permitted as a tenant isolation backstop — this is security enforcement, not business logic.
5	Every ResultHeader and ResultValue MUST be traceable to exactly one HL7_Message.
6	Unmapped codes quarantine — never auto-guess, never silently discard.

2. HL7 Ingestion Layer

The ingestion layer receives raw HL7 ORU messages, archives them, and feeds the normalization pipeline. This is the entry point for all lab data in Dx7.

2.1 HL7_Message

Raw HL7 message archive. Every downstream record traces back to exactly one row here.

Field	Type / Constraint	Notes
`hl7_message_id`	PK	Message identifier
`tenant_id`	FK → Tenant	Receiving tenant

`message_control_id`	string	MSH-10 value
`raw_payload`	text/blob	Complete HL7 message as received — retained for traceability
`received_at`	timestamp	Ingestion timestamp
`processed_flag`	boolean	Successfully parsed and stored
`quarantine_flag`	boolean	Contains unmapped or invalid data

⚠ raw_payload must be retained encrypted at rest. Never discard original HL7.

⚠ UNIQUE INDEX on (tenant_id, message_control_id). Protects against HL7 retransmission duplicates — engines resend on missed ACKs.

2.2 Transport

HCLab transmits HL7 via MLLP (real-time push, confirmed). MLLP is unencrypted TCP — direct transmission over public internet is prohibited. Secure transport topology is being finalized with Arnel (LABExpress IT). A file-based folder watch may be used as a temporary development workaround only — it is not the production transport path.

Source	HCLab (LABExpress LIS)
Format	HL7 ORU^R01
Transmission Mode	Real-time MLLP push (confirmed)
Option A: VPN Tunnel	HCLab → VPN/WireGuard → MLLP Listener on DigitalOcean → Ingestion Module
Option B: Edge Relay	HCLab → MLLP → Edge Relay (on-prem, stateless) → HTTPS → DigitalOcean Ingestion Module
Dev Workaround Only	File-based folder watch permitted for local development and testing. NOT production transport.
Security	No unencrypted HL7 over public internet. Option A or B required for production.

3. Identity Layer

Dx7 v1 uses a simplified identity model consistent with SantéXA CDM v1.0. Full GlobalPatient/TenantPatientMap is implemented at canonical platform transition. v1 uses tenant-local patient identity only.

3.1 Tenant

Field	Type / Constraint	Notes
`tenant_id`	PK	Unique tenant identifier
`name`	string	e.g., LABExpress
`code`	string, UNIQUE	Short identifier
`logo_url`	string, nullable	PL branding
`footer_text`	string, nullable	e.g., Powered by LABExpress
`primary_color`	string, nullable	Hex color for accent

`is_active`	boolean	Active status
`created_at`	timestamp	

3.2 Client (Dialysis Clinic)

Field	Type / Constraint	Notes
`client_id`	PK	Unique clinic identifier
`tenant_id`	FK → Tenant	Parent tenant
`name`	string	Clinic display name
`code`	string	Maps to HL7 location code
`logo_url`	string, nullable	Clinic branding
`address`	text, nullable	
`is_active`	boolean	
`created_at`	timestamp	

3.3 Patient

Patients are created automatically from HL7. No manual patient creation in Dx7. Corrections must be made in LIS.

Field	Type / Constraint	Notes
`patient_id`	PK	Dx7 patient identifier
`tenant_id`	FK → Tenant	
`client_id`	FK → Client	
`lis_patient_id`	string	PID-3 from HL7 — LIS source of truth
`name`	string	PID-5 from HL7
`birthdate`	date, nullable	PID-7 from HL7
`gender`	enum M|F|O, nullable	PID-8 from HL7
`is_active`	boolean	Soft delete only
`created_at`	timestamp	

⚠ UNIQUE INDEX on (tenant_id, client_id, lis_patient_id). Duplicate detection on ingestion.

4. Order & Result Layer

4.1 Order

Field	Type / Constraint	Notes
`order_id`	PK	

tenant_id	FK → Tenant	
client_id	FK → Client	Referring clinic
patient_id	FK → Patient	
accession_number	string	OBR-3 value
source_hl7_message_id	FK → HL7_Message	Provenance — required
released_at	timestamp, nullable	Result release timestamp
created_at	timestamp	

4.2 ResultHeader

Test-level result record grouping analyte values under one ordered test.

Field	Type / Constraint	Notes
result_id	PK	
order_id	FK → Order	
tenant_id	FK → Tenant	
source_hl7_message_id	FK → HL7_Message	Provenance — required
sxa_test_id	FK → SXA_Test_Catalog	Canonical test identity — from OBR-4 mapping
specimen_type	string, nullable	Specimen description
collection_datetime	timestamp, nullable	Specimen collection time
result_datetime	timestamp, nullable	Result generation time

4.3 ResultValue

Individual analyte measurement. One row per OBX segment. Stores display value, numeric value, unit, reference range, and abnormal flag as pass-through from lab.

Field	Type / Constraint	Notes
result_value_id	PK	
result_id	FK → ResultHeader	
tenant_id	FK → Tenant	Required — global rule #1
analyte_code	FK → SXA_Analyte	Canonical analyte — from OBX-3 mapping
display_value	string	Value as displayed to nurse
value_numeric	decimal, nullable	Parsed numeric value
unit	string, nullable	OBX-6 as received
reference_range_low	decimal, nullable	Lab-provided low bound — pass-through
reference_range_high	decimal, nullable	Lab-provided high bound — pass-through

abnormal_flag	string, nullable	Raw OBX-8 only — H, L, or blank. Pass-through. No interpretation.
raw_hl7_segment	text	Complete OBX segment — retained for traceability

⚠ abnormal_flag is raw pass-through from OBX-8. Dx7 displays it as-is. No clinical interpretation is applied.

5. Test Catalog (SXA)

SantéXA-native canonical test identity layer. Dx7 maps all HCLab OBX-3 and OBR-4 codes to this catalog. The catalog is owned by the Product Owner — do not add analytes without authorization.

5.1 SXA_Test_Catalog

Field	Type / Constraint	Notes
sxa_test_id	PK	e.g., SXA_TEST_BUN_PRE
canonical_name	string	e.g., Blood Urea Nitrogen (Pre-Dialysis)
category	string	e.g., Chemistry, Hematology
result_type	enum SINGLE|PANEL	
active_flag	boolean	

5.2 SXA_Analyte

Field	Type / Constraint	Notes
analyte_code	PK	e.g., SXA_A_BUN_PRE
display_name	string	e.g., BUN Pre-Dialysis
default_unit	string, nullable	e.g., mg/dL
result_type	enum NUMERIC|TEXT|ENUM	

5.3 SXA_Analyte Seed Data — Dx7 Dialysis Scope

Complete analyte catalog for Dx7 v1. BUN PRE and POST are defined as distinct analytes. This is a Dx7 extension to SantéXA CDM v1.0 — the parent CDM carries only SXA_A_BUN (singular). The split is required for dialysis adequacy monitoring.

analyte_code	display_name	default_unit	result_type
SXA_A_HGB	Hemoglobin	g/dL	NUMERIC
SXA_A_HCT	Hematocrit	%	NUMERIC
SXA_A_WBC	White Blood Cells	$x10^3/\mu L$	NUMERIC
SXA_A_PLT	Platelets	$x10^3/\mu L$	NUMERIC
SXA_A_BUN_PRE	**BUN Pre-Dialysis**	mg/dL	NUMERIC

SXA_A_BUN_POST	**BUN Post-Dialysis**	mg/dL	NUMERIC
SXA_A_CREAT	Creatinine	µmol/L	NUMERIC
SXA_A_NA	Sodium	mEq/L	NUMERIC
SXA_A_K	Potassium	mEq/L	NUMERIC
SXA_A_CA	Calcium	mg/dL	NUMERIC
SXA_A_PHOS	Phosphorus	mg/dL	NUMERIC
SXA_A_ALB	Albumin	g/dL	NUMERIC
SXA_A_GLU	Glucose (FBS)	mg/dL	NUMERIC

→ SXA_A_BUN_PRE and SXA_A_BUN_POST are highlighted. These are Dx7 dialysis extensions. Parent SantéXA CDM v1.0 carries SXA_A_BUN (singular) — this split will be reconciled at platform transition.

6. Normalization Mapping (SXA Map)

Deterministic mapping of HCLab HL7 codes to canonical SXA identifiers. No auto-guessing. No fuzzy matching. If a code cannot resolve, it quarantines.

6.1 OBR-4 → SXA_Test_Catalog (Tenant_Test_Map)

Resolves HL7 OBR-4 Universal Service ID to canonical test identity. Store tenant_test_code as identifier component before first caret.

tenant_test_code (OBR-4)	sxa_test_id	Notes
DGC0074	SXA_TEST_CBC	Complete Blood Count
DGC0035	SXA_TEST_FBS	Fasting Blood Sugar
BUNPRE	SXA_TEST_BUN_PRE	Pre-Dialysis BUN order
BUNPOST	SXA_TEST_BUN_POST	Post-Dialysis BUN order — confirm OBR-4 code with Arnel
K	SXA_TEST_K	Potassium

⚠ BUNPOST OBR-4 code must be confirmed with Arnel (LABExpress IT) before build. Placeholder shown above.

6.2 OBX-3 → SXA_Analyte (Tenant_Analyte_Map)

Resolves HL7 OBX-3 Observation Identifier to canonical analyte identity. One row per distinct OBX-3 code emitted by HCLab.

tenant_analyte_code (OBX-3)	analyte_code	Notes
BUNPRE	SXA_A_BUN_PRE	Confirmed in HL7 samples
BUNPOST	SXA_A_BUN_POST	Confirm OBX-3 code with Arnel

K	SXA_A_K	Confirmed in HL7 samples
HGB / HCT / WBC / PLT	*SXA_A_HGB / HCT / WBC / PLT*	*OBX-3 codes not in current samples — will quarantine until confirmed*
NA / CL / CO2 / CA / CREAT	*SXA_A_NA / CL / CO2 / CA / CREAT*	*OBX-3 codes not in current samples — will quarantine until confirmed*

→ Grey rows are pending — unmapped OBX-3 codes will correctly quarantine at analyte level per Section 7 until confirmed codes are added.

6.3 Quarantine Behavior

Condition	Behavior
OBR-4 cannot resolve	Set HL7_Message.quarantine_flag = TRUE. Do not create ResultHeader.
OBX-3 cannot resolve	Quarantine that ResultValue only. Other mapped analytes in same ResultHeader persist.
Auto-guessing	PROHIBITED. No fuzzy matching. No silent acceptance.
Silent discard	PROHIBITED. Every failure must be logged and flagged.

7. Dx7 Operational Tables

These tables are specific to Dx7 workflow and do not exist in the SantéXA CDM. They are the permanent strategic asset of the Dx7 build.

7.1 Session

Created when a patient is selected for a shift. One session per patient per shift per day.

Field	Type / Constraint	Notes
`session_id`	PK	UUID
`tenant_id`	FK → Tenant	
`client_id`	FK → Client	
`patient_id`	FK → Patient	
`session_date`	date	
`shift_number`	integer 1–4	
`chair`	string, nullable	Free text — number or letter
`assigned_by`	FK → User	Who created the session
`assigned_at`	timestamp	
`created_at`	timestamp	

⚠ UNIQUE INDEX on (patient_id, session_date, shift_number). One session per patient per shift.

7.2 Chair_Audit

Field	Type / Constraint	Notes
`id`	PK	
`tenant_id`	FK → Tenant	Required — global rule #1
`session_id`	FK → Session	
`chair_old`	string, nullable	Previous assignment
`chair_new`	string	New assignment
`changed_by`	FK → User	
`changed_at`	timestamp	

7.3 MD_Notes

Field	Type / Constraint	Notes
`id`	PK	
`tenant_id`	FK → Tenant	
`session_id`	FK → Session	Per session, not per patient
`md_user_id`	FK → User	MD role only
`note_text`	text	Free text
`created_at`	timestamp	
`updated_at`	timestamp	Editable within 24 hours by MD only

8. Users & Auth

Field	Type / Constraint	Notes
`user_id`	PK	
`tenant_id`	FK → Tenant	
`client_id`	FK → Client, nullable	Null for PL Admin
`email`	string, UNIQUE	Login credential
`password_hash`	string	
`name`	string	
`role`	enum	sysad | pl_admin | clinic_admin | charge_nurse | shift_nurse | md
`is_active`	boolean	
`created_at`	timestamp	

⚠ JWT tokens — 8-hour expiry (shift length). tenant_id enforced on ALL queries. No auth tokens in localStorage or sessionStorage.

9. Traceability Chain

Every result displayed in Dx7 must be traceable through the following chain. If any link is missing, the record is invalid.

Step	Entity	Key Link
1	HL7_Message	Raw HL7 file received and archived
2	Order	source_hl7_message_id → HL7_Message
3	ResultHeader	order_id → Order, source_hl7_message_id → HL7_Message
4	ResultValue	result_id → ResultHeader, analyte_code → SXA_Analyte
5	Display	Dx7 API reads ResultValue, presents display_value as pass-through. No transformation.